

COURSE TITLE : **BIO INFORMATICS**
COURSE CODE : **5242**
COURSE CATEGORY : **E**
PERIODS/WEEK : **4**
PERIODS/SEMESTER : **52**
CREDITS : **4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Molecular Biology and Genomic Databases	13
2	Sequence alignment	13
3	Genomic data processing	13
4	Gene structure in Prokaryotes and Eukaryotes	13
TOTAL		52

Course General Outcome :

MODULE	G.O	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE:
1	1	To understand the basic concept of molecular biology
	2	To understand Genomic Databases
2	3	To understand sequence alignments
	4	To understand multiple sequence alignments
	5	To know phylogenetic analysis
3	6	To understand Genomic data and its processing
4	7	To understand Gene structure in Prokaryotes and Eukaryotes:

G.O.: General Outcome

On completion of the study the student will be able:

MODULE I: MOLECULAR BIOLOGY AND GENOMIC DATABASES

1.1.0 To understand the basic concept of molecular biology

- 1.1.1 To define cells, Chromosomes, DNA, RNA and Proteins
- 1.1.2 To explain types of DNA and RNA
- 1.1.3 To describe the Central dogma of molecular biology

1.2.0 To understand Genomic Databases

- 1.2.1 To define Genomes and Genes
- 1.2.2 To define Genetic code, Transcription, Translation and Protein synthesis.
- 1.2.3 To explain Web based genomic and proteomic data bases: NCBI, Gen Bank

MODULE II: SEQUENCE ALIGNMENT

2.1.0 To understand sequence alignments

- 2.1.1 To explain Dot Plot and Pair-wise sequence alignments.
- 2.1.2 To explain local sequence similarity
- 2.1.3 To explain global sequence similarity
- 2.1.4 To define distance measures in sequence alignments.
- 2.1.5 To state the concept of Smith-Waterman algorithm

2.2.0 To understand multiple sequence alignment

- 2.2.1 To explain multiple sequence alignments
- 2.2.2 To state the concept of sum-of-pairs measure
- 2.2.3 To explain star and tree alignments in multiple sequence alignment
- 2.2.4 To differentiate between PAM and BLOSUM

2.3.0 To know phylogenetic analysis

- 2.3.1 To describe phylogenetic analysis

MODULE III: GENOMIC DATA PROCESSING

3.1.0 To understand Genomic data and its processing

- 3.1.1 To state the concept of Informational view of Genomic data.
- 3.1.2 To explain Genomic Signal Processing
- 3.1.3 To describe DNA spectrograms
- 3.1.4 To state the concept of Identification of protein coding regions
- 3.1.5 To explain gene expression
- 3.1.6 To state the concept of microarrays.
- 3.1.7 To state the concept of microarray image analysis

MODULE IV: GENE STRUCTURE IN PROKARYOTES AND EUKARYOTES

4.1.0 To understand Gene structure in Prokaryotes and Eukaryotes

- 4.1.1 To state the basic concepts and terminologies related to molecular structures
- 4.1.2 To define basic molecular Visualization
- 4.1.3 To explain RNA secondary structure prediction
- 4.1.4 To define Protein folding
- 4.1.5 To explain Protein folding problem
- 4.1.6 To state the concept of Protein Threading
- 4.1.7 To define Protein Visualization
- 4.1.8 To state the concept of drug discovery
- 4.1.9 To state the concept of BLAST for analyzing sequences

CONTENTS

MODULE I MOLECULAR BIOLOGY AND GENOMIC DATABASES

Basic Concepts of Molecular Biology – Definition of Cells, Chromosomes, DNA, RNA and Proteins - Central dogma of molecular biology

Genomic Databases

Definition of Genomes, Genes, Genetic code, Transcription, Translation and Protein synthesis - Web based genomic and proteomic data bases - NCBI, Gen Bank

MODULE II: SEQUENCE ALIGNMENT

Dot plot-Pair-wise sequence alignments-local and global-Sequence similarity and distance measures - Smith-Waterman algorithm

Multiple sequence alignment –Sum-of-Pairs measure -Star and tree alignments –PAM and BLOSUM - Phylogenetic analysis

MODULE III: GENOMIC DATA PROCESSING

Informational view of Genomic data - Genomic Signal Processing - DNA Spectrograms - Identification of protein coding regions - Gene expression - Microarrays - Microarray image analysis

MODULE IV: GENE STRUCTURE IN PROKARYOTES AND EUKARYOTES

Gene structure in Prokaryotes and Eukaryotes - Molecular Structure Prediction - Basic concepts and terminologies related to molecular structures - Basic molecular Visualization - RNA secondary structure prediction - Protein folding problem – definition of Protein Threading - Protein Visualization – Basic concept of Drug Discovery.

Case Study Software Tools - Use of Tools for basic and specialized sequence processing - BLAST

TEXT BOOKS

1. Bioinformatics -A Beginners Guide - Claverie & Notredame Wiley-Dreamtech India Pvt Ltd, 2003.
2. Introduction to Bioinformatics - Lesk, Oxford University Press, Indian Edition, 2003

REFERENCES

1. Introduction to Computational Molecular Biology - Setubal & Meidanis,Thomson: Brooks/Cole, International Student Edition, 2003
2. Bioinformatics:Sequence, structure and databanks - Higgins and Taylor,Oxford University Press, Indian Edition, 2003
3. Bioinformatics Computing - Bergeron, Prentice hall of India, 2003
4. Current topics in Computational Molecular Biology - Jiang, Xu and Zhang, Ane Books, New Delhi, 2004
5. Bioinformatics method and application Genomics,Proteinomics & drug discovery - S.C Rastogi & Namitha Mendiratta,
6. Microarray Bioinformatics - Dov Stekel, Cambridge University Press